

tf.compat.v1.linalg.diag_part Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/linalg/diag_part

Returns the batched diagonal part of a batched tensor.

Function Signatures

```
tf.compat.v1.linalg.diag_part( input, name='diag_part', k=0, padding_value=0, align='RIGHT_LEFT' )
```

Code Examples

```
tf.compat.v1.linalg.diag_part(  
    input,  
    name='diag_part',  
    k=0,  
    padding_value=0,  
    align='RIGHT_LEFT'  
)
```

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input,
name='diag_part',
k=0,
padding_value=0,
align='RIGHT_LEFT'
)
```

```
diagonal[i, j, ..., l, n]
= input[i, j, ..., l, n+y, n+x] ; if 0 <= n+y < M and 0 <= n+x < N,
padding_value ; otherwise.
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padding_value ; otherwise.
```

```
offset = max_diag_len - diag_len(d) ; if (`align` in {RIGHT_LEFT,
RIGHT_RIGHT}
and `d >= 0`) or
(`align` in {LEFT_RIGHT, RIGHT_RIGHT}
and `d <= 0`)
0 ; otherwise
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offset = max_diag_len - diag_len(d) ; if (`align` in {RIGHT_LEFT,
RIGHT_RIGHT}
and `d >= 0`) or
(`align` in {LEFT_RIGHT, RIGHT_RIGHT}
and `d <= 0`)
0 ; otherwise

```

Documentation

Returns the batched diagonal part of a batched tensor.

Returns a tensor with the $k[0]$ -th to $k[1]$ -th diagonals of the batched input.

Assume input has r dimensions $[I, J, \dots, L, M, N]$.

Let max_diag_len be the maximum length among all diagonals to be extracted,
 $\text{max_diag_len} = \min(M + \min(k[1], 0), N + \min(-k[0], 0))$

Let num_diags be the number of diagonals to extract,
 $\text{num_diags} = k[1] - k[0] + 1$.

If $\text{num_diags} == 1$, the output tensor is of rank $r - 1$ with shape
 $[I, J, \dots, L, \text{max_diag_len}]$ and values:

where $y = \max(-k[1], 0)$, $x = \max(k[1], 0)$.

Otherwise, the output tensor has rank r with dimensions
 $[I, J, \dots, L, \text{num_diags}, \text{max_diag_len}]$ with values:

where $d = k[1] - m$, $y = \max(-d, 0) - \text{offset}$, and $x = \max(d, 0) - \text{offset}$.

offset is zero except when the alignment of the diagonal is to the right.

where $\text{diag_len}(d) = \min(\text{cols} - \max(d, 0), \text{rows} + \min(d, 0))$.

The input must be at least a matrix.

For example:

Returns

Raises